

Math 514, F19
Exam 2

Name: *Answer*

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	24	
2	12	
3	18	
4	14	
5	14	
6	18	
Total	100	

Good Luck !

Problem 1. (24 points)

(a) Assume the nominal interest rate is 10%. What are the effective interest rates respectively when compounded monthly and daily?

$$\begin{aligned} r &= 0.1 \\ \text{when compounded monthly} \\ r_{\text{eff}} &= \left(1 + \frac{r}{12}\right)^{12} - 1 = \left(1 + \frac{0.1}{12}\right)^{12} - 1 \approx 0.1047 \\ \text{when compounded daily} \\ r_{\text{eff}} &= \left(1 + \frac{r}{365}\right)^{365} - 1 = \left(1 + \frac{0.1}{365}\right)^{365} - 1 \approx 0.1052 \end{aligned}$$

(b) The income stream $\mathbf{a} = (1000, 1500, 2500, 4000)$ consists of payments made at the end of the next four years. Find the present value of $\mathbf{a}$ for a nominal interest rate of 9% compounded continuously.

$$\begin{aligned} r &= 0.09 \\ PV(\mathbf{a}) &= 1000e^{-r} + 1500e^{-2r} + 2500e^{-3r} + 4000e^{-4r} \\ &= 1000e^{-0.09} + 1500e^{-0.18} + 2500e^{-0.27} + 4000e^{-0.36} \\ &\approx 6865.99 \end{aligned}$$

(c) A loan for \$100,000 to be paid back over 180 months at a nominal interest rate of 6% compounded monthly, what would be the monthly payment?

$$\begin{aligned} r &= \frac{0.06}{12} = 0.005 \quad \text{interest rate per month} \\ \frac{A}{1+r} + \frac{A}{(1+r)^2} + \dots + \frac{A}{(1+r)^{180}} &= 100,000 \\ \text{let } \beta &= \frac{1}{1+r} = \frac{1}{1.005}, \text{ then} \\ A\beta + A\beta^2 + \dots + A\beta^{180} &= 100,000 \\ A\beta \frac{1-\beta^{180}}{1-\beta} &= 100,000 \\ \Rightarrow A &= 100,000 \cdot \frac{1-\beta}{\beta(1-\beta^{180})} \approx 843.46 \end{aligned}$$

Problem 2. (12 points) A certain investment costs \$3400. The investment produces income of \$1500 at the end of the first year and \$2500 at the end of the second year. Find the annual rate of return on the investment.

$$\begin{aligned}
 3400 &= \frac{1500}{1+r} + \frac{2500}{(1+r)^2} \\
 \text{let } \beta &= \frac{1}{1+r}, \text{ then} \\
 2500\beta^2 + 1500\beta - 3400 &= 0 \\
 \Rightarrow 25\beta^2 + 15\beta - 34 &= 0 \\
 \Rightarrow \beta &= \frac{-15 \pm \sqrt{15^2 - 4 \times 25 \times (-34)}}{2 \times 25} = \frac{-15 + \sqrt{3625}}{50} \\
 &\approx 0.9042 \Rightarrow r = \frac{1}{\beta} - 1 \approx 1.106 - 1 = 0.106
 \end{aligned}$$

Problem 3. (18 points) Assuming that the nominal interest rate is 4.8%, compounded continuously.

(a) A stock trades today at \$68.32, find a theoretical lower bound for the price of a call option with strike price $K = \$67.50$ expiring two months from now.

By the put-call option parity formula,

$$\begin{aligned}
 S + P - C &= Ke^{-rt} \\
 C &= S + P - Ke^{-rt} \\
 &\geq S - Ke^{-rt} \quad (\text{since } P \geq 0) \\
 &= 68.32 - 67.50 e^{0.048/6} \\
 &\approx 1.358
 \end{aligned}$$

(b) Suppose that the no-arbitrage price of a call option (with strike price and time to expiration as in (a)) is \$1.95. Find the no-arbitrage price for a European put option with the same strike price and expiration date.

$$\begin{aligned}
 S + P - C &= Ke^{-rt} \\
 \Rightarrow P &= Ke^{-rt} + C - S \\
 &= 67.50 e^{-0.048/6} + 1.95 - 68.32 \\
 &\approx 0.592
 \end{aligned}$$

Problem 4. (14 points) Argue that if $S + P - C < Ke^{-rt}$, then there is an arbitrage by providing a sure-win strategy.

An arbitrage is obtained by:

★ buying the security, buying the put and selling the call.

Initial cost: $S + P - C$

Return at time t : K

The present value of the gain is $Ke^{-rt} - (S + P - C) > 0$.

Problem 5. (14 points) An experiment can result in any of the outcomes 1, 2, or 3. If there are two different wagers with

$$r_1(1) = 2, r_1(2) = 4, r_1(3) = -9,$$

$$r_2(1) = 6, r_2(2) = 12, r_2(3) = -10,$$

Is an arbitrage possible?

We need to check whether there exists probabilities (P_1, P_2, P_3)

$(P_i \geq 0, i=1,2,3)$ such that

$$\begin{cases} P_1 r_1(1) + P_2 r_1(2) + P_3 r_1(3) = 0 \\ P_1 r_2(1) + P_2 r_2(2) + P_3 r_2(3) = 0 \\ P_1 + P_2 + P_3 = 1 \end{cases} \Rightarrow \begin{cases} 2P_1 + 4P_2 - 9P_3 = 0 \\ 6P_1 + 12P_2 - 10P_3 = 0 \\ P_1 + P_2 + P_3 = 1 \Rightarrow P_3 = 1 - P_1 - P_2 \end{cases}$$

$$\Rightarrow \begin{cases} 11P_1 + 13P_2 = 9 \\ 16P_1 + 22P_2 = 10 \end{cases} \Rightarrow \begin{matrix} P_1 = 2 \\ P_2 = -1 \end{matrix} \quad \text{and thus } P_3 = 0$$

but $P_2 = -1 < 0$ is impossible, thus an arbitrage is possible.

Problem 6. (18 points) Suppose that the price of a security either goes up by a factor of $3/2$ or goes down by a factor of $1/2$ in one period. We also assume the periodically compounded nominal interest rate is $r = 10\%$ and the initial price of the security is 100.

(a) Compute the risk-neutral probabilities $(p^*, 1 - p^*)$.

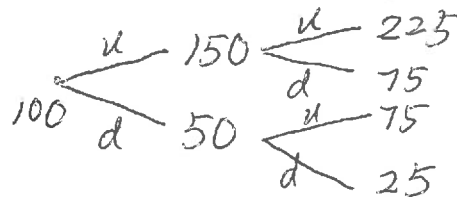
$$u = 3/2, d = 1/2, r = 0.1.$$

$$p^* = \frac{1+r-d}{u-d} = \frac{1+0.1-0.5}{1.5-0.5} = \frac{0.6}{1} = 0.6$$

$$1-p^* = 1-0.6 = 0.4$$

(b) Determine the no-arbitrage cost of a **European put** option to sell the security at the end of two periods for a strike price of 125.

$$S = 100, t = 2, K = 125$$



$$P_{uu} = 0$$

$$P_{ud} = P_{du} = (125 - 75)(1+r)^{-2} = \frac{50}{(1.1)^2} \approx 41.32$$

$$P_{dd} = (125 - 25)(1+r)^{-2} = \frac{100}{(1.1)^2} \approx 82.64$$

Thus

$$\begin{aligned}
 P &= P_{uu}(p^*)^2 + P_{ud}p^*(1-p^*) + P_{du}(1-p^*)p^* + P_{dd}(1-p^*)^2 \\
 &= 0 + 41.32 \times 2 \times (0.4 \times 0.6) + 82.64 \times (0.4)^2 \\
 &\approx 33.06
 \end{aligned}$$